RAJALAKSHMI ENGINEERING COLLEGE

TAMILSELVAN P 251801349-2025-AIDS

T2

Started on	Tuesday, 25 August 2026, 11:45 PM
State	Finished
Completed on	Monday, 31 August 2026, 11:56 AM
Time taken	5 days 12 hours
Marks	0.00/1.00
Grade	**0.00** out of 10.00 (**0**%)

Question 1 | Incorrect Mark 0.00 out of 1.00

A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the person needs to run a distance to burn out his calories.

If he has eaten i burgers with c calories each, then he has to run at least $3^i * c$ kilometers to burn out the calories. For example, if he ate 3 burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are $(3^0 * 1) + (3^1 * 3) + (3^2 * 2) = 1 + 9 + 18 = 28$.

But this is not the minimum, so need to try out other orders of consumption and choose the minimum value. Determine the minimum distance he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm.Apply greedy approach to solve the problem.

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Input Format

First Line contains the number of burgers

Second line contains calories of each burger which is n space-separate integers

Output Format

Print: Minimum number of kilometers needed to run to burn out the calories

Sample Input

```
3
5 10 7
```

Sample Output

```
76
```

For example:

Test	Input	Result
Test Case 1	3 1 3 2	18

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int cmpDesc(const void *a, const void *b) {
    int x = *(const int *)a;
    int y = *(const int *)b;
    return y - x;  // descending order
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) {
        return 0;
    }

    if (n <= 0) {
        printf("0\n");
        return 0;
    }

    int *cal = (int *)malloc(n * sizeof(int));
```

```c
    if (cal == NULL) {
        return 1;
    }

    for (int i = 0; i < n; i++) {
        scanf("%d", &cal[i]);
    }

    qsort(cal, n, sizeof(int), cmpDesc);

    long long totalDistance = 0;
    long long power = 1; // 3^0

    for (int i = 0; i < n; i++) {
        totalDistance += (long long)cal[i] * power;
        power *= 3;
    }

    printf("%lld\n", totalDistance);

    free(cal);
    return 0;
}
```

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	Test	Input	Expected	Got	
✔	Test Case 1	3 1 3 2	18	18	✔
✔	Test Case 3	3 5 10 7	76	76	✔

Your code failed one or more hidden tests.

Your code must pass all tests to earn any marks. Try again.

Incorrect

Marks for this submission: 0.00/1.00.

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